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# KARTIK BHATIA

College Park, MD

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## EDUCATION

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University of Maryland

College Park, MD

B.S., Aerospace Engineering, GPA: 3.3

May 2027

Relevant Coursework: Structural Analysis; Control Systems; Space Navigation and Guidance; Vibration & Aeroelasticity; Engineering lab I & II

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## SKILLS

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Engineering - Requirements Management, Risk Management (FMEA), Systems Engineering Documentation (N<sup>2</sup> Diagrams, ConOps, V&V Plans, Trade Studies), Design Reviews, Environmental Test Planning, Stakeholder Communication, Schedule & Budget Management

Fabrication - FDM, SLA, SLS, Powder Bed Fusion (3D Printing), 3D Scanning, DMG Mori 5-axis CNC, Drill Mill, Lathe, Water Jet, Laser Cutting

Software - Fusion 360, Siemens NX, MATLAB, Python, C++, SolidWorks, ROS2, IBM DOORS, Jira, CFD (Ansys Fluent), Ansys STK, Geomagic X

Robotics & ML - ROS2, Gazebo, NVIDIA Jetson, PyTorch, OpenCV, Edge Inference

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## SUMMARY

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Aerospace engineering student with **hands-on** experience in space **systems integration**, environmental qualification, **hardware verification** and launch campaign logistics. Led end-to-end readiness for a 12U-class payload flight, coordinating **subsystem integration**, **test sequencing**, and launch day operations. Bring a systems engineering mindset to delivering reliable hardware in demanding environments, on schedule.

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## EXPERIENCE

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Terrapin Works - Advanced Fabrication Lab

College Park, MD

Additive Manufacturing Technician

February 2026 - Present

- **Operate** across **FDM, SLA, SLS, Multi Jet Fusion, powder bed fusion and composites 3D printers**, handling slicing, setup, part harvesting
- Perform **dimensional verification** and **3D scanning** using the **Arctec Leo** handheld scanner and the **Hexagon ROMER Arm** for part conformance
- Maintain and **troubleshoot** fabrication **equipment**, oversee scheduled upkeep, and enforce safety SOPs to ensure consistent lab operations

Collaborative Controls and Robotics Lab

College Park, MD

Undergraduate Research Assistant

August 2025 - Present

- Overhauled **2/6 Khepera IV** robots, tracing test run dropouts to a shared network channel and loose motor connectors, **restoring clean test runs**
- Implemented decentralized task allocation nodes in **simulation** runs; completed tasks **~15% faster** than the initial centralized baseline
- **Built** a **Python** based **automated data logging pipeline** for physics based simulation trials to enable repeatable **KPI tracking** across **test runs**

V.A.A.C.E - NASA L'SPACE Program

College Park, MD

Chief Engineer

August 2024 - Present

- **Lead** a **\$20K NASA seed-funded** research project with **22+** members, building a **scientific instrument payload** and an **autonomous sensor driven inspection and repair system** toward a **TRL-4** prototype
- Designed the complete **payload assembly** in **SolidWorks** and **3D-printed** the corresponding sensor housing and structural components
- Directed **electronics assembly, Arduino/ESP32 firmware development**, and **integration testing** across subteams for flight readiness
- Facilitated subsystem **FMEAs**, identifying failure modes and **mitigating high severity risk** exposure by **~65%**
- Led the **flight operations** for the **12U-class** payload's launch to **90,000 ft**, **validating hardware in near-space** conditions
- Leading development of an **autonomous inspection and repair system**, architecting design in SolidWorks and iterating on the **control logic**

Space Copy Inc.

Houston, TX

Robotics Mechanical Engineering Intern

June 2025 - August 2025

- Mapped interactions between a **7-DOF robotic arm** and an **industrial 3D printer**, documenting pick-and-place sequences and homing routines
  - Designed a **magnet-based end-effector assembly** in **SolidWorks**, **validating 1100 N grip strength**, communicating trade-offs to **stakeholders**
  - Built a physics based model in **Gazebo**, **simulating** tool pick-and-place cycles, validating full workspace **coverage within ± 30 inches**
  - Performed **CFD** simulations in **Ansys Fluent** on **melted regolith** in microgravity, characterizing melt-pool stability at **1,600 K**
  - Surveyed **20+ lunar** sites via **LROC imagery**, ranking **Ilmenite/Anorthite** regions by solar exposure and extraction feasibility
  - Characterized **thermal noise** in **Raman spectra** of **heated regolith**, defining correction requirements for **material detection**
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## PROJECTS

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Zero Pressure Aerobot - UMD Balloon Payload Program

College Park, MD

Prototyping Engineer

September 2024 - December 2024

- Engineered a **high-altitude balloon** system for **near-space payload testing**, reaching **60,000 ft** while carrying a **4 lb payload** in a full scale flight
- **Designed** and 3D-printed **structural components** that held **<2 % strain** during ascent, **extending flight duration by ~25 %**
- **Programmed** an **Arduino**-based **cut-down mechanism** with a **fail-safe logic**, enabling **payload recovery** within a **200 m landing radius**

Terraformers - URC Rover Team

College Park, MD

Fabrication and Testing Engineer

January 2024 – June 2024

- Conducted **CNC** machining and **welding** for a **7 ft × 4 ft rover** build, aligning fabrication execution to **±0.5 mm tolerance** requirements
- Owned **validation** trials and execution reporting, confirming **5 m/s traversal over 2 ft obstacles across ~50 km** with **zero structural failures**
- **Reinforced chassis** plates, **reducing** predicted **bending stress by ~35 %** and **vibration amplitude by ~25 %** under impact loads